Experiment 3

Constructing a DC Power Supply

SAFETY

Make sure that you have read the **General Safety Notes**, in the Introductory section of this manual, before you begin.

Do not, **under any circumstances** attempt to repair any of the equipment. If you suspect equipment to be faulty, turn it off at the power point and talk to your demonstrator.

CAPACITORS: ONE WAY ONLY!

The capacitors you will use in this exercise are electrolytic capacitors, so it is **essential** that they are wired in circuits the correct way around. If connected incorrectly, they may become **irreparably damaged** or even explode.

Connect the black '–' end to the earth side of the circuit, as shown in the circuit diagrams below. Double-check all connections before turning on the power. If in doubt, talk to your demonstrator.

ALWAYS TURN OFF THE POWER!

Before adding or removing any element from your circuit, make sure to turn off any power or signal source. Aside from the risk of shock, having too much current running through some components can cause them to be **irreparably damaged**, or even catch fire. If in doubt, talk to your demonstrator.

DIODES: DO NOT CONNECT DIRECTLY ACROSS THE POWER SUPPLY!

Never connect a 'bare' diode directly across a power supply or signal source. A large current will flow through the diode and it will most likely **destroy** the diode. **Always** make sure that there is a resistor wired in series with any diode.

If you see smoke or detect an unpleasant smell, like that of plastic burning, **immediately** turn off the power and ask your demonstrator for assistance.

Outline of Laboratory Exercise

In this lab you will be investigating how to make a DC power supply from an AC source. The two important components which make up rectification circuits are **capacitors and diodes**. Diodes are the electrical analog of a valve as they only allow the current in a single direction. Capacitors on the other hand are devices which can store charge. Each device is important in the process of converting AC to DC known as **Rectification.**

✓ ***Pre-lab exercises****: Read the entire laboratory exercise and Sections D, E and F of the Appendices, then complete the online prelab questions. Pay close attention to the sections on Oscilloscopes.*

Learning Goals

- To further develop electronic skills of wiring and adding to circuits.
- To develop an understanding of how various electronic components such as diodes and capacitors alter the current and voltage in a circuit.
- To proficiently use a digital CRO application to observe and interpret the resulting signal.
- To create a DC Power supply.

Introduction

Before we begin you will familiarize yourself with the device known as a Cathode Ray Oscilloscope (CRO). The link http://hyperphysics.phy-astr.gsu.edu/hbase/electronic/scopec.html will give a description of the function of each of the controls on the CRO.

Use the NI Elvis Oscilloscope program for the CRO and initially set the Signal Generator to produce a **square wave** at a frequency of **200 Hz**. Adjust the CRO until you can see the wave pattern. Note: the signal generator can produce several different wave shapes; make sure yours is a square wave.

A capacitor commonly consists of two parallel conductors with a non-conducting region between them. A capacitor can store charge on the plates when a voltage is applied. The charge Q stored on each of the plates of a capacitor of capacitance C, is always related to the potential difference V between the plates by:

$$Q = CV$$

The capacitors in a rectification circuit are used to flatten the signal making it more constant in value. This is done by allowing the capacitor to discharge while no voltage is supplied.

Part A *Capacitors*

Given the rate of change of charge calculated above, the charge on the discharging capacitor decreases exponentially with time, i.e.

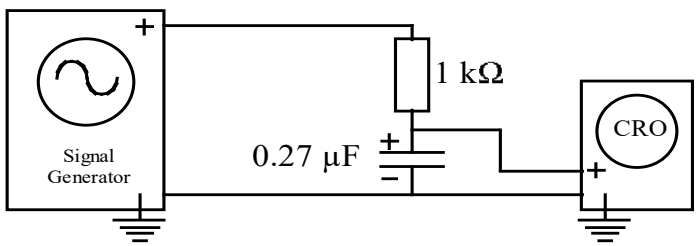
$$\mathrm{q}(t) = Q_0 e^{-\frac{t}{RC}} = Q_0 e^{-\frac{t}{\tau}}.$$

The capacitor will therefore reach 37% (or 1/e) of its initial charge after a characteristic time

$$\tau = RC$$

The value τ is known as the **time constant** of the RC (resistor-capacitor) combination.

To investigate the discharge of a capacitor set up the following circuit.



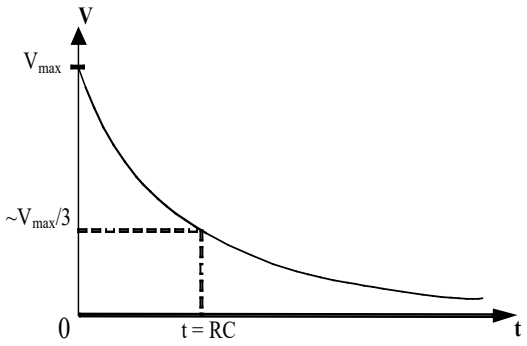
Set the signal generator to supply a 3 V peak-to-peak **square wave** at a frequency of **200 Hz**. You should confirm this signal is as you expect by connecting the CRO directly across the signal generator output. Check with your demonstrator that your CRO is set up correctly.

NOTE: Your CRO is (probably) **not** broken! It is very easy to make a mistake configuring the circuit, the signal generator or the CRO settings, so that you cannot see the correct signal. Double-check that all settings and connections are correct before you give up. As your demonstrator for help if you can't get things to work as they should.

Once you have confirmed that the signal generator is producing the correct square wave, connect the capacitor and resistor in the circuit and observe the charge/discharge curve. Include a sketch of what you see on the CRO display in your log notebook.

Use the discharge curve, measure the time constant of the circuit, which will be the time taken for the voltage to drop to 37% of the difference between the maximum and minimum voltage values on the CRO display. Compare your measured value to the expected value of the time constant.

Example of Discharge Curve



Next, replace the $1\ k\Omega$ resistor with a $10\ k\Omega$ resistor.

How is the discharge curve different from the curve for the 1 kΩ resistor? Discuss and compare the properties of each curve. Try reducing the **frequency** of the signal to 20 Hz. What difference does that make? Comment.

Now measure the time constant of the RC circuit with the $10\ k\Omega$ and compare it to the expected value.

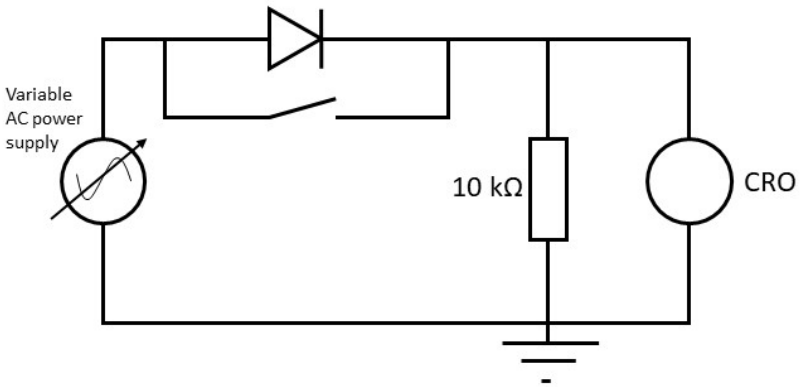
Part B *Diodes*

You will now investigate the properties of diodes by looking at their properties. A good description of the diode's inner workings can be found in Appendix G and the link here: http://hyperphysics.phy-astr.gsu.edu/hbase/solids/diode.html

A diode only allows current to flow through it in one direction, making it a useful component for the purpose of rectification. Rectifying an AC current means making it flow in one direction only.

NOTE: Be especially careful in the following sections to disconnect or turn off the voltage source before modifying the circuit. Connecting a diode across a power supply *without* including a resistor in series with the diode will most likely destroy the diode, as a very large current will flow across it.

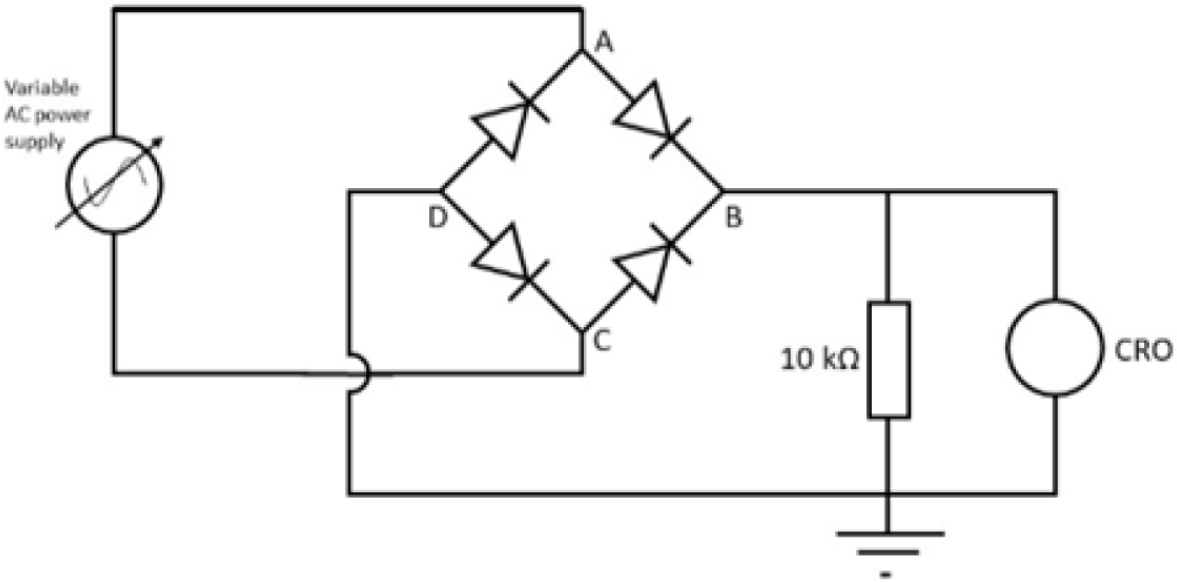
Set up the circuit below. For the 'Variable AC Power supply', it is preferable to use the signal generator, set to produce a **sine wave** at **20** Hz. Alternatively, an AC power supply set to **6 V** can be used, which will produce an approximately sinusoidal signal at 50 Hz. Before turning on the circuit, please have it checked by your demonstrator.



Observe the signal on the CRO. Compare the signal with the switch open and closed. Explain what you see.

The circuit above is an example of a **half-wave rectifier**. The 'problem' with this circuit is that power is only delivered to the resistor (load) for half the time during each cycle of the AC power supply.

With some clever thinking we can create a circuit that allows for a constant flow of current, but in only one direction. This is known as a **full-wave rectifier**:



The power supply on the left outputs a sinusoidal AC current.

Copy this circuit diagram into your logbook. On the circuit diagram, trace the path of current flow in the circuit for the case where Point A is at a higher potential than point C. Then, using a different colour ink, trace the path of current flow when point C is at a higher potential than A.

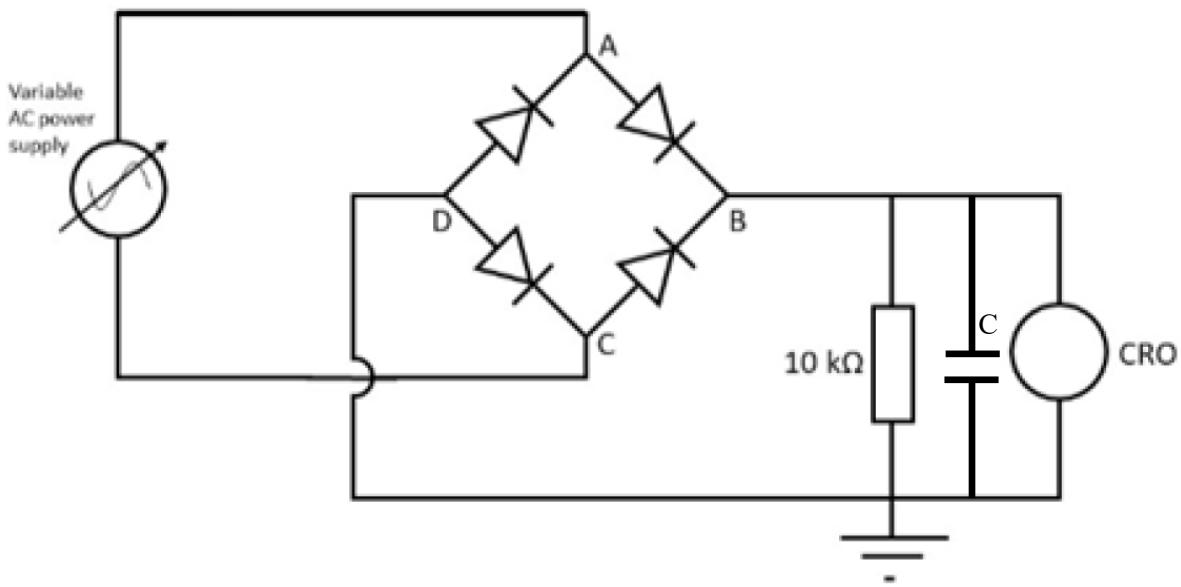
Record your prediction of what you think the measured signal will look like on the CRO.

Now connect the circuit and compare your observations to what you predicted.

Part C *AC/DC Smoothing*

Although diodes can rectify an AC signal, rectification alone does not provide a **constant** DC voltage or current. To smooth out the wavy 'bumps' of the rectified signal, we need something more.

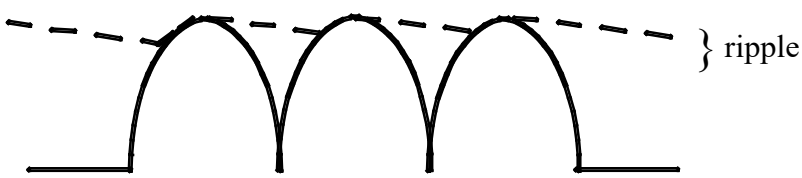
Add a $0.27\ \mu\mathrm{F}$ capacitor into the full-wave rectification circuit, as shown:



What is the effect of the capacitor on the rectified output. Do this for three different capacitors (on your desk there should be a board containing capacitors of 0.27 µF, 4.0 µF and 100 µF capacitance).

Analysis

- What is happening? Explain what the capacitors are doing and how they achieve this effect.
- Sketch the observed signals to scale, superimposing the smoothed waves over the rectified output (use the diagram below as an example). Carefully consider the actual peak voltage values.



- If possible, record the peak-to-peak voltage range of the 'ripple' signal for each of the three capacitors.

Conclusion

Summarise your findings and how well they compare with the theory.